

Multi-epoch Growth Rate Test of EFC Perturbation-Sector Coupling

$f\sigma_8(z)$ from BOSS DR12, eBOSS DR16, and DESI DR1

Morten Magnusson

ORCID: [0009-0002-4860-5095](https://orcid.org/0009-0002-4860-5095)

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Abstract. We present the first multi-epoch test of the Energy-Flow Cosmology (EFC) perturbation-sector coupling using 14 measurements of the growth rate $f\sigma_8(z)$ spanning $z = 0.067$ – 1.491 from three generations of spectroscopic surveys: 6dFGS, BOSS DR12, eBOSS DR16, and DESI DR1 (full-shape and peculiar velocities). We fit a regime-dependent gravitational modification $\mu(a) = 1 - B \cdot T(a)$ with gate function $T(a) = [1 + (a_t/a)^n]^{-1}$ ($n = 2$ fixed), integrating the full linear growth ODE. Λ CDM provides an excellent fit: $\chi^2 = 12.07/12$ dof ($\Omega_m = 0.302$, $\sigma_8 = 0.774$). EFC yields $\chi^2 = 11.98/11$ dof with $B = 0.078$ ($\mu_0 = 0.922$), giving $\Delta\chi^2 = 0.10$ (0.06σ , 2 extra parameters). $B = 0$ (Λ CDM) lies within the 1σ profile interval. The data do not require modified growth; EFC is consistent but not preferred. We identify an Ω_m – B degeneracy that prevents $f\sigma_8$ data alone from constraining the transition redshift z_t . Cross-probe combination with BAO is required to break this degeneracy.

1 Introduction

The growth rate of cosmic structure, parameterised by $f\sigma_8(z)$, provides a direct test of gravitational physics at cosmological scales. In Energy-Flow Cosmology (EFC), the effective gravitational coupling μ deviates from unity due to entropy-stiffness effects in the perturbation sector. The EFC perturbation survival valley, mapped by systematic CMB+LSS localisation [1], requires $\mu_0 \in [0.93, 0.96]$ with $\Sigma \in [1.03, 1.07]$ and gravitational slip $\eta \neq 1$.

A pilot analysis [2] using only BOSS DR12 (3 data points at $z = 0.38, 0.51, 0.61$) with a constant coupling α_{L2} found a mild 1.7σ preference for $\alpha_{L2} > 0$, but also identified a sign conflict between BAO and RSD channels. The WP1a reference model [3] demonstrated that a scale-free gate function with $\mu_0 = 0.85$ closes 73% of the S_8 gap.

This note extends the pilot in three ways: (i) 14 data points spanning $z = 0.067$ – 1.491 from four survey programmes, (ii) regime-dependent $\mu(a)$ via the gate function rather than a constant coupling, and (iii) full numerical integration of the growth ODE.

2 Data

We compile 14 measurements of $f\sigma_8(z)$ from published analyses. All errors are treated as independent (diagonal covariance). Cross-survey correlations between overlapping samples (e.g., BOSS $z = 0.51$ and DESI LRGz0 $z = 0.510$) are not accounted for; this is conservative for $\Delta\chi^2$ comparisons.

Table 1: Multi-epoch $f\sigma_8(z)$ compilation.

Survey	Tracer	z_{eff}	$f\sigma_8 \pm \sigma$	Reference
6dFGS	Galaxies	0.067	0.423 ± 0.055	Beutler+ 2012
DESI DR1 PV	FP+TF	0.070	0.450 ± 0.055	Turner+ 2025
DESI DR1 BGS	BGS	0.295	0.408 ± 0.040	DESI 2024 V
BOSS DR12	LOWZ	0.380	0.497 ± 0.045	Alam+ 2017
BOSS DR12	CMASS	0.510	0.458 ± 0.038	Alam+ 2017
DESI DR1	LRGz0	0.510	0.449 ± 0.032	DESI 2024 V
BOSS DR12	CMASS	0.610	0.436 ± 0.034	Alam+ 2017
eBOSS DR16	LRG	0.700	0.473 ± 0.041	Gil-Marín+ 2020
DESI DR1	LRGz1	0.706	0.413 ± 0.029	DESI 2024 V
eBOSS DR16	ELG	0.850	0.315 ± 0.095	de Mattia+ 2021
DESI DR1	LRG+ELG	0.930	0.381 ± 0.034	DESI 2024 V
DESI DR1	ELG	1.317	0.396 ± 0.062	DESI 2024 V
eBOSS DR16	QSO	1.480	0.462 ± 0.045	Neveux+ 2020
DESI DR1	QSO	1.491	0.318 ± 0.076	DESI 2024 V

3 Method

3.1 Growth equation with modified gravity

We solve the linear growth ODE in scale factor a :

$$D'' + \frac{3}{2a} [1 + \Omega_{\text{DE}}(a)] D' = \frac{3}{2a^2} \Omega_m(a) \mu(a) D \quad (1)$$

where $\Omega_m(a) = \Omega_{m,0} a^{-3}/E^2(a)$, $\Omega_{\text{DE}}(a) = (1 - \Omega_{m,0})/E^2(a)$, and $E^2(a) = \Omega_{m,0} a^{-3} + (1 - \Omega_{m,0})$ for flat Λ CDM background. Primes denote d/da .

The EFC modification enters through:

$$\mu(a) = 1 - B \cdot T(a), \quad T(a) = \frac{1}{1 + (a_t/a)^n} \quad (2)$$

with $a_t = 1/(1 + z_t)$ and gate width $n = 2$ fixed (WP1a reference). At $z = 0$: $\mu_0 = 1 - B$. Λ CDM corresponds to $B = 0$.

Initial conditions: $D(a_i) = a_i$, $D'(a_i) = 1$ at $a_i = 10^{-3}$ (matter-dominated growing mode). The growth rate is $f(z) = d \ln D / d \ln a = (a/D) D'$, and $f\sigma_8(z) = f(z) \cdot \sigma_{8,0} \cdot D(z)/D(0)$.

3.2 Fitting strategy

Λ CDM has 2 free parameters $(\Omega_m, \sigma_{8,0})$. For EFC, we fix Ω_m at the Λ CDM best-fit value and fit $(\sigma_{8,0}, B, z_t)$ — 3 parameters. This choice is motivated by the Ω_m – B degeneracy: with Ω_m free, the optimizer drifts to $\Omega_m \approx 0.45$, $B \approx 0.48$ (boundary), compensating higher matter density with stronger growth suppression. Fixing Ω_m breaks this degeneracy and isolates the growth-sector signal.

We use `differential_evolution` (SciPy) for global optimisation, followed by local polishing via L-BFGS-B. The profile likelihood for B is computed by minimising over $(\sigma_{8,0}, z_t)$ at each fixed B value.

4 Results

4.1 Best-fit parameters

Table 2: Best-fit parameters and model comparison.

Parameter	Λ CDM	EFC ($n = 2$)
Ω_m	0.3019	0.3019 (fixed)
$\sigma_{8,0}$	0.7739	0.7916
B	—	0.078
z_t	—	3.00 (boundary)
μ_0	1 (by def.)	0.922
χ^2	12.07	11.98
dof	12	11
χ^2/dof	1.006	1.089
$\Delta\chi^2$	0.095 (0.06 σ)	
ΔAIC	−1.90 (Λ CDM pref.)	

4.2 Profile likelihood for B

The profile likelihood (Figure 1, lower-left) shows a broad, nearly flat minimum centred near $B \approx 0.08$. The 1σ interval extends from $B = 0$ to $B \approx 0.40$, meaning Λ CDM ($B = 0$) is fully consistent with the data. The survival valley ($\mu_0 \in [0.93, 0.96]$, $B \in [0.04, 0.07]$) lies within the 1σ contour but is not preferentially selected by the data.

4.3 Key finding: Ω_m – B degeneracy

With Ω_m free, the EFC fit drifts to $\Omega_m = 0.45$, $B = 0.48$ ($\mu_0 = 0.52$, far outside any physically motivated range). This demonstrates that $f\sigma_8$ data alone cannot simultaneously constrain Ω_m and B : higher Ω_m increases the matter-dominated growth, which is then suppressed by larger B , producing a nearly degenerate valley in parameter space. Breaking this degeneracy requires external geometric constraints (BAO, SN Ia, or CMB distance priors).

5 Discussion

5.1 Interpretation

The multi-epoch $f\sigma_8(z)$ data are fully consistent with Λ CDM ($\chi^2/\text{dof} = 1.006$). EFC with $\mu(a) < 1$ provides no statistically significant improvement ($\Delta\chi^2 = 0.10$, $\Delta\text{AIC} = -1.9$ favouring Λ CDM). This is a null result for the growth sector: the data do not require modified gravitational coupling.

However, this does *not* exclude EFC. The survival valley $\mu_0 \in [0.93, 0.96]$ lies well within the 1σ profile contour. A modification at the 4–7% level in μ produces growth-rate changes of order 1–3% in $f\sigma_8$, which is below the current measurement precision (~ 5 –15% per point). The data lack the statistical power to distinguish the valley from Λ CDM.

5.2 Comparison with prior results

The pilot analysis [2] found $\alpha_{L2} = 0.34$ (1.7σ) using BOSS DR12 alone. That result used a constant coupling and only 3 data points. With the extended dataset and regime-dependent $\mu(a)$, the preference weakens to $< 0.1\sigma$. This is expected: the constant α_{L2} conflated BAO and RSD signals; the regime-dependent approach correctly separates background geometry from perturbation growth.

The WP1a reference model ($B = 0.187$, $\mu_0 = 0.813$) predicts a systematically lower $f\sigma_8(z)$ trajectory (Figure 1, dashed red), which undershoots the data at $z < 0.5$. The best-fit $B = 0.078$ is milder, consistent with the survival valley edge.

5.3 Limitations

1. **Diagonal covariance:** Cross-correlations between surveys sharing sky overlap (BOSS/eBOSS, BOSS/DESI at $z \sim 0.5$) are not included. A full analysis requires the joint covariance matrix.
2. Ω_m **fixed:** The nested model comparison is conditional on $\Omega_m = 0.302$. A joint BAO+ $f\sigma_8$ fit with free Ω_m is the definitive test.
3. z_t **unconstrained:** The transition redshift hits the prior boundary ($z_t = 3.0$), indicating the data do not constrain when the modification turns on. This is consistent with a late-time activation that is smooth enough to be absorbed by σ_8 .
4. **Scale-independent μ :** We use $\mu(a)$ without k -dependence. The stiffness mechanism $R \propto k^{-4}$ [4] predicts scale-dependent effects that would require full-shape $P(k)$ fitting.

5.4 Next steps

The RSD Full Analysis (planned in the EFC validation ledger as “Multi-epoch RSD growth trajectory”) requires: joint BAO+ $f\sigma_8$ likelihood with full covariance (breaking the Ω_m – B degeneracy); DESI DR2 full-shape data (when released); and scale-dependent $\mu(k, a)$ from the EFC relativistic action [5].

6 Reproducibility

All code, data, and results are provided as supplementary files: `efc_multi_epoch_v2.py` (analysis script), `efc_multi_epoch_results.json` (numerical output), `efc_multi_epoch_results.png` (four-panel diagnostic plot). The analysis requires only NumPy, SciPy, and Matplotlib.

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